Business Plan for Decision Tree Classification Assignment

EventEdge Corporate Training: Predicting Participant Dropout Risk

Prepared for a Decision Tree model and business interpretation assignment

Business name	EventEdge Corporate Training
Business area	B2B corporate training and workforce development
Business problem	Identify participants who are likely to drop out before completing a paid training course.
Machine learning task	Binary classification using a Decision Tree model
Target variable	High_Dropout_Risk: Yes or No
Dataset file	eventedge_training_dropout_risk_dataset.xlsx

1. Executive Summary

EventEdge Corporate Training is a B2B learning company that sells professional development programs to organizations. Its courses include Data Analytics, Leadership, Project Management, Customer Experience, Digital Marketing, and AI for Business. Clients pay for employees to participate, expecting stronger skills, higher productivity, and measurable learning outcomes.

The main business challenge is participant dropout. When employees stop attending or fail to complete a course, EventEdge risks lower client satisfaction, reduced renewal rates, wasted coaching capacity, and weaker training outcomes. The company wants to use early engagement data from the first week to predict which participants may need extra support.

2. Business Model and Value Proposition

- EventEdge sells online and hybrid corporate training programs to client companies.
- Revenue comes from course fees paid per participant or through company-level training packages.
- The company differentiates itself through personalized learning support, progress monitoring, and business-focused course design.
- Client companies expect high completion rates because training completion is connected to employee development goals.

3. Business Problem

Participant dropout creates operational and financial problems. A participant who disengages early may still occupy a seat, receive reminders, and require administrative follow-up. If many participants fail to complete training, client companies may question the value of the program and may not renew their contracts.

The business needs an early warning system that can answer this question:

Which enrolled participants are at high risk of dropping out, based on early engagement and business context?

4. How Machine Learning Can Help

A Decision Tree classification model can help EventEdge classify each participant as either high dropout risk or not high dropout risk. The model can support decisions such as:

- Which participants should receive an early check-in from a coach or program advisor.
- Which client companies may need a manager-support reminder.
- Which course types may require redesign, shorter modules, or additional onboarding support.
- How to prioritize limited support resources without treating every participant the same.

5. Dataset and Target Variable

The dataset represents synthetic participant-level records from EventEdge training programs. Each row is one participant enrolled in a course. The target variable is:

High_Dropout_Risk

Positive class: Yes = the participant is at high risk of dropping out before course completion. **Negative class:** No = the participant is not classified as high dropout risk.

6. Main Input Features

Feature group	Example columns	Business meaning
Client context	Industry, Company_Size	Shows the type and scale of the client organization.
Participant context	Job_Level, Workload_Level	Indicates how work demands and role seniority may affect completion.
Course context	Course_Type, Course_Length_Weeks, Fee_Paid	Shows the type, duration, and revenue value of the training.
Early engagement	Sessions_Attended_First_Week, LMS_Logins_First_Week, Assignment1_Submitted	Measures whether the participant is active at the beginning of the course.
Support signals	Manager_Support_Level, Reminder_Emails_Opened, Satisfaction_Score_Week1	Shows motivation, organizational support, and early satisfaction.

7. Business Interpretation of Prediction Outcomes

Prediction outcome	Meaning in this business	Possible business action
True Positive	The model correctly identifies a participant who is at high dropout risk.	Offer coaching, reminders, and manager follow-up early.
True Negative	The model correctly identifies a participant who is not at high dropout risk.	Continue normal course support and monitoring.
False Positive	The model predicts high dropout risk, but the participant would have completed the course.	Extra support may be unnecessary, but usually the cost is moderate.
False Negative	The model misses a participant who is actually at high dropout risk.	This is risky because the company may fail to intervene before the participant disengages.

8. Most Important Evaluation Metric

Recall for the positive class can be especially important. In this business case, missing a participant who is truly at high dropout risk can be costly because EventEdge may lose the opportunity to intervene. A high recall means the model identifies many of the participants who need support. However, precision should also be reviewed because too many false positives can waste coaching time.

9. How Feature Importance Can Help the Business

- If Assignment1_Submitted is highly important, EventEdge should monitor first-assignment completion very closely.
- If LMS_Logins_First_Week is highly important, low platform activity can become an early warning signal.
- If Workload_Level is highly important, the company may need flexible deadlines or manager involvement for busy employees.
- If Manager_Support_Level is highly important, EventEdge can advise client companies to actively support participants during the course.
- If Course_Length_Weeks is highly important, long courses may need milestone-based support or modular redesign.

10. Limitations, Bias, and Responsible Use

The dataset is synthetic and simplified. It may not capture all real-world causes of dropout, such as personal circumstances, organizational culture, workload changes, illness, or manager communication quality. Some features may also reflect structural differences between companies rather than individual motivation.

- The model should not be used to blame employees for disengagement.
- Predictions should be reviewed by human program managers before action is taken.
- Support actions should be helpful and respectful, such as reminders, coaching, deadline flexibility, or manager communication.
- The model should be monitored over time to ensure it does not systematically disadvantage specific groups, industries, or job levels.

11. Student Guidance for the Assignment

Students should connect their technical results to the business plan. The notebook should not only report model metrics; it should explain what those metrics mean for EventEdge.

- Use High_Dropout_Risk as the target variable.
- Do not use Participant_ID as an input feature.
- Check and handle missing values before model training.

- Compare training accuracy and testing accuracy to discuss overfitting.
- Interpret feature importance in practical business language.
- Explain false positives and false negatives using the corporate training context.